

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: Total Marks: Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I _YAGGOLUPENCHALAKONA(192312599) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

AI in Agriculture: Precision Farming and Crop Monitoring

Artificial Intelligence (AI) is transforming modern agriculture by enabling **precision farming, crop-health monitoring, disease detection, pest identification, crop-yield prediction, and efficient resource management**. The reviewed research evaluates **Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), and Convolutional Neural Network (CNN)** techniques. Important input parameters include **soil moisture, temperature, humidity, rainfall, pest density, crop-stress index, satellite imagery, UAV/multispectral imagery, weather data, and soil information**. IoT field sensors provide real-time environmental and soil information, while preprocessing involves missing-value treatment, outlier handling, normalization, feature scaling, and image augmentation.

The experimental results demonstrate the strong potential of AI for precision agriculture. **CNN achieved the highest accuracy of 95.6% and F1-score of 94.6%**, making it highly effective for image-based crop and disease analysis. Random Forest achieved **91.7% accuracy** and provided a good balance between performance and computational cost, while Decision Tree was more computationally efficient. The research also reports that a hybrid crop-yield prediction model using meteorological and soil data achieved an **R² value of 0.88**. Overall, integrating **AI with IoT sensors, UAVs, satellite imagery, and remote sensing** can enable continuous crop monitoring, early disease and pest detection, improved yield prediction, and optimized agricultural resources.



Revolutionizing agriculture: 5G-enabled IoT networks for smart farming

Nadeem Yousuf Khanday

Computational intelligence techniques for 5G enabled IoT networks, 191-212, 2025

The integration of 5G technology with the Internet of Things (IoT) is poised to revolutionize modern agriculture, creating a paradigm shift toward precision farming and sustainable food production. The primary goal of our study is to highlight the transformative potential of 5G-enabled IoT networks in the context of smart farming, where high-speed connectivity, ultra-low latency, and massive device-to-device communication capabilities offer unprecedented opportunities for data-driven agricultural practices. 5G networks enhance the efficiency of IoT sensors, drones, autonomous machinery, and real-time analytics platforms, enabling seamless monitoring of crops, soil, and environmental conditions at unprecedented scales and granularities. By facilitating real-time decision-making, predictive analytics, and automation, 5G-enabled IoT networks support resource optimization, yield maximization, and the minimization of environmental impacts. Furthermore, the study illustrates the challenges related to infrastructure deployment, data security, and the digital divide in rural areas. The findings underscore the necessity for strategic investments in 5G infrastructure and regulatory frameworks to fully harness the potential of smart farming technologies, thereby paving the way for a more sustainable, resilient, and technologically advanced agricultural sector. Finally, the case study is presented which fuses technologies like 5G-enabled IoT networks, devices, and deep learning models to offer transformative potential for the agricultural sector. As these technologies continue to evolve, their fusion not only helps in addressing major challenges but also enables precise, efficient, and sustainable farming practices, ultimately leading to higher productivity and profitability. The case study mainly focuses on crop disease detection using hierarchical modeling and successfully achieving better results.